

# NICHOLAS WOOD

## Staff Software Engineer

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## SUMMARY

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- Staff Software Engineer with over nine years of experience designing large-scale, **AI-driven platforms** across healthcare and enterprise domains.
- Skilled in building **multi-tenant, event-driven systems** using **FastAPI**, **Node.js**, and **TypeScript**, with deep expertise in **AWS**, **Kafka**, and **PostgreSQL**.
- Hands-on experience integrating **LLM** orchestration frameworks like **LangChain**, **LangGraph**, and **MCP** into production workflows to automate operations and improve efficiency.
- Strong background in **EHR** and **FHIR** integrations, **HIPAA-compliant** architectures, and secure cloud infrastructure for AI and data-intensive applications.
- Known for leading **cross-functional teams**, **mentoring** engineers, and delivering reliable, scalable software that bridges modern AI with real- world business impact.

## EDUCATION

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**Missouri University of Science and Technology,** 07/2014 – 07/2016 | Rolla, MO  
*Master's in Computer Science Degree*

**Missouri University of Science and Technology,** 09/2010 – 06/2014 | Rolla, MO  
*Bachelor's in Computer Science Degree*

## EXPERIENCE

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**Notable, Staff Software Engineer** 📍 08/2022 – 07/2025 | San Mateo, CA  
*An AI platform for healthcare operations that automates patient-access, clinical, and revenue-cycle workflows across major EHRs. Think agentic workflows + low/no-code configuration + deep EHR/payer integrations.*

- Helped design and scale Notable's AI automation platform, integrating event-driven **FastAPI** + **Node.js** + **Golang** microservices, **LLM orchestration** with **LangGraph**, vector search (**pgvector/Pinecone**), and streaming inference endpoints-powering over a million autonomous agent workflows per day across enterprise healthcare systems.
- Led the build-out of multi-tenant event-driven services in **FastAPI** and **Node/TypeScript**, connected to major EHR systems (Epic, Cerner, Athena) through secure FHIR and HL7 integrations.
- Built internal **MCP**-based connectors so AI agents could talk directly to systems like Outlook, Salesforce, CRMs and insurance APIs without human intervention.

- Worked closely with the AI/ML team to deploy **LangChain** and **LangGraph**-based orchestration flows, powering conversational and reasoning agents that summarize claims, check benefits, and assist with clinical documentation.
- Introduced vector-search pipelines (**pgvector** + **Pinecone**) to match unstructured documents and automate prior-authorization lookups across patient records.
- Designed secure tokenization and data isolation strategies for PHI and user-level audit trails to maintain full HIPAA and SOC 2 compliance.
- Set up streaming and observability pipelines (**Kafka**, **Postgres**, **S3**, **CloudWatch**) to track automation latency, error patterns, and ROI metrics across thousands of care sites.
- Designed and managed **AWS**-based infrastructure supporting LLM orchestration, vector search, and event-driven APIs - leveraging **ECS Fargate**, **Lambda**, **S3**, **CloudWatch**, and **RDS (Postgres)** for scalable, HIPAA-compliant operations.
- Conducted periodic security and compliance audits (**IAM** policies, encryption at rest, network isolation) to maintain **SOC 2 / HIPAA** certification standards.
- Built real-time data interfaces for **AI agents** and automation workflows using **SSE**, **WebSockets**, and **React Query**, ensuring instant feedback loops between clinicians and back-office systems.
- Collaborated with product designers to develop an AI activity dashboard visualizing automation states, token usage, and workflow metrics with **Recharts**, **D3.js**, and **Grafana** panels.
- Partnered with clinicians and operations leads to translate manual workflows into AI-first agent flows, cutting repetitive administrative work by 60%+ and speeding up patient intake cycles.
- Led a small staff group of backend and ML engineers, mentoring them on service boundaries, structured retries, and **CI/CD** with **AWS CDK**.
- Built internal tools and dashboards in **Looker** and **Grafana** to monitor agent success rates, token usage, and cost metrics for **LLM** inference.
- Mentored cross-functional engineers in test-driven development (**TDD**), scalable service design, and best practices for deploying reliable, production-grade AI systems.

## SKILLS

AI/ML, LLM/AI System, Gen AI, NLP, BERT, RAG, FAISS, Ray, LlamaIndex, Hugging Face Transformer, Software Architecture, Software Design, LangChain, Langgraph, Fine Tuning, MCP, A2A, HL7 v2, FHIR, EHR APIs, Python/FastAPI, Pytorch, Node.js, TypeScript, Golang, ETL/ELT, MLOps, React, Next.js, PostgreSQL, Snowflake, Redis, S3, dbt/Airflow ETL, Amazon Bedrock, Kafka, SQS/SNS, JWT/OAuth2, Docker, Github, Terraform, CDK, ECS/Fargate, CloudWatch

**BitsWits**, Senior Software Engineer

06/2018 – 06/2022 | Huston, TX

*Product engineering/consulting building large-scale web & mobile apps for enterprise and startup clients; heavy on cloud-native backends, real-time UX, and AWS deployments.*

- Led the end-to-end design and development of large-scale web and mobile applications for enterprise and startup clients, using **React**, **Nextjs**, **Node.js**, **Golang** and **Python (FastAPI)** backed by **PostgreSQL** and **Redis**.
- Collaborated directly with product and client teams to translate requirements into scalable architecture diagrams, **GraphQL** APIs, and **AWS**- based deployment pipelines (**ECS**, **Lambda**, **S3**, **CloudFront**).
- Designed event-driven microservices supporting analytics, notifications, and behavior tracking, powered by **Kafka**, **Redis Streams**, and **GraphQL** federation gateways.

- Engineered real-time communication layers (**WebSockets**, **Server-Sent Events**, and **WebRTC**) for live dashboards, chat, and telemetry-reducing latency and improving engagement by 0%+
- Unified front-end codebases by introducing a shared **React / React Native** component library and monorepo tooling (**Nx**, **Turborepo**), cutting UI development cycles by 25%+
- Established TypeScript, Jest, ESLint, and Cypress as engineering standards, driving adoption of TDD and consistent coding practices across multiple product teams.
- Implemented **CI/CD** pipelines with GitHub Actions and Docker, automating integration tests, staging previews, and zero-downtime deployments.
- Integrated third-party APIs (Stripe, Firebase, Twilio, Google Maps) and built secure authentication flows using **JWT**, **OAuth 2.0**, and **AWS Cognito** for multi-tenant applications.
- Mentored engineers on clean code, TDD workflows, peer review discipline, and observability best practices, helping standardize performance and reliability across projects.
- Participated in architecture reviews, hiring panels, and internal governance, shaping BitsWits' engineering standards for scalability, testing, and security.

## SKILLS

**React**, **Next.js**, **React Native**, **GraphQL**, **Node.js**, **Golang**, **Python (FastAPI)**, event-driven microservices, **PostgreSQL**, **Redis**, **Kafka**, **Redis Streams**, **AWS (ECS, Lambda, S3, CloudFront)**, **Nx/Turborepo**, **TypeScript**, **Jest**, **ESLint**, **Cypress**, **GitHub Actions**, **Docker**, **JWT**, **OAuth2**, **AWS Cognito**, **Stripe**, **Firebase**, **Twilio**, **Google Maps**

**Oracle Health**, *Junior Developer* 

01/2017 – 05/2018 | Kansas City, MO

**Healthcare IT/EHR platforms powering patient scheduling, orders, documentation, and integrations.**

- Shipped features and fixes across **Java/Spring** services backed by **Oracle Database** (SQL/PL/SQL), exposing REST endpoints used by patient scheduling, orders, and documentation modules.
- Built HL7/FHIR adapters for internal services (ADT, orders, results), adding validation and error routing that reduced integration failure tickets by ~20%.
- Wrote unit/integration tests with **JUnit/Mockito** and added lightweight contract tests for downstream consumers; raised coverage on a core service from ~5%~70%.
- Helped migrate legacy **SOAP** endpoints to **REST** with clearer resource models, request validation, and standardized error responses.
- Contributed to **CI** with **Jenkins** (later **GitHub Enterprise**), adding static analysis (SpotBugs/Checkstyle), test gates, and artifact versioning; reduced release rollbacks during quarterly drops.
- Containerized dev/test environments with **Docker**, created reproducible local stacks, and documented runbooks that sped up onboarding for new engineers.
- Implemented PHI-safe audit logging, encryption-in-transit (mTLS), and role-based access checks, aligning services with HIPAA and internal compliance guidelines.
- Collaborated with analysts and QA to triage production issues, reproduce edge cases, and deliver hotfixes with clear rollback plans and change records.

## SKILLS

Java, Spring, SQL/PLSQL, REST design, SOAP→REST migration, error handling/validation, HL7 v2, FHIR adapters & validation, JUnit, Mockito, contract testing, coverage improvement, Jenkins, GitHub Enterprise, static analysis (SpotBugs/Checkstyle), Dockerized local/dev environments, mTLS, RBAC, PHI-safe audit logging, HIPAA-aligned controls

Oracle Health, *Engineer Intern*

07/2016 – 12/2016 | Kansas City, MO

## SKILLS

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### AI/ML

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LLM Development & Fine-Tuning, RAG, Langchain, Langgraph, Langbase, LangSmith, Hugging Face, Autogen, CrewAI, DSPy, Open AI, Llama 3, PyTorch, TensorFlow, Ray, MLflow, Pinecone, pgvector, FAISS, Multi agent system, Prompt Engineering & Evaluation, SageMaker, GCP Vertex AI, CUDA

### Languages & Runtime

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JavaScript (ES6+), TypeScript, Python, Node.js, Golang, Java (JVM), C# (.NET), C++, Rust, SQL, Bash/Shell/PowerShell, YAML/JSON/Protobuf, GraphQL, CUDA, Julia, WebAssembly(WASM), Ray/Dask, Hugging Face Inference/OpenAI/vLLM runtimes

### Backend & API

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Microservices & Distributed Systems, Rest, GraphQL, gRPC, Node.js (Express, NestJS), Python (Flask, FastAPI, Django), Golang(Gin, Fiber), .NET Core/Java Spring Boot, OAuth2, JWT, SSO, RBAC/ABAC, Prisma, TypeORM, SQLAlchemy, EF, Gorm, Redis, Kafka, RabbitMQ, AWS SQS, Websocket, WebRTC, SSE, Pub/Sub, AWS API gateway, NGINX, Kong, Cloudflare

### DevOps

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Terraform, AWS CDK, CloudFormation, Ansible, AWS (ECS, Fargate, Lambda, S3, RDS, CloudWatch, EventBridge), GCP & Azure, CI/CD, Github Actions, Jenkins, GitLab CI, Docker, Kubernetes, Prometheus/Grafana, ELK, OpenTelemetry, CloudWatch, SLO/SLA/SLI Design, VPC & Networking, VPN, Zero-Trust, Auto-Scaling

### Frontend

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JavaScript, TypeScript, React/Next.js, Redux, React Query, SWR, TTI, LCP, CLS, SSR, SSG, SEO, Tailwind CSS, Styled-Component, MUI, Figma, Motion, D3.js, Chart.js, Webpack, Vite, Angular, Vue

### Testing

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TDD, BDD, Pytest/unittest (Python), Jest/Mocha/Vitest, xUnit/Nunit, Cypress, Selenium, Playwright, Model Evaluation Pipelines, Prompt Evaluation, Pandera, k6, Locust, JMeter, OWSAP ZAP